

Data Flow Diagram

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to illustrate how weather and rainfall data flows through the Flood Prediction System. The diagram should represent the interaction between users, the machine learning prediction process, datasets, trained models, and the prediction results. It shows how input data is processed, stored, analyzed, and transformed into flood risk predictions for disaster management and early warning.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (<i>e.g., Meteorologist, Disaster Management Authority, Resident/User</i>).
Rectangle with numbered header	Process	An activity that transforms incoming data into useful output (<i>e.g., Collect Weather Data, Preprocess Data, Train ML Model, Predict Flood Risk</i>).
Rectangle (solid fill, no number)	Data Store	A place where data is stored for future use (<i>e.g., Historical Rainfall Dataset, Trained Machine Learning Model, Prediction Database</i>).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores (<i>e.g., Weather Data, Processed Data, Prediction Request, Flood Prediction Result</i>).

Example

The Data Flow Diagram below illustrates the movement of weather and rainfall data through the

Flood Prediction System, beginning with user input, followed by data preprocessing, machine learning-based flood prediction, and finally displaying the flood risk prediction to the user.

Data Flow Diagram – Flood Prediction System



